

Risk factors for ischaemic and intracerebral haemorrhagic stroke in 22 countries (the INTERSTROKE study): a case-control study.

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BACKGROUND: The contribution of various risk factors to the burden of stroke worldwide is unknown, particularly in countries of low and middle income. We aimed to establish the association of known and emerging risk factors with stroke and its primary subtypes, assess the contribution of these risk factors to the burden of stroke, and explore the differences between risk factors for stroke and myocardial infarction. **METHODS:** We undertook a standardised case-control study in 22 countries worldwide between March 1, 2007, and April 23, 2010. Cases were patients with acute first stroke (within 5 days of symptoms onset and 72 h of hospital admission). Controls had no history of stroke, and were matched with cases for age and sex. All participants completed a structured questionnaire and a physical examination, and most provided blood and urine samples. We calculated odds ratios (ORs) and population-attributable risks (PARs) for the association of all stroke, ischaemic stroke, and intracerebral haemorrhagic stroke with selected risk factors. **FINDINGS:** In the first 3000 cases (n=2337, 78%, with ischaemic stroke; n=663, 22%, with intracerebral haemorrhagic stroke) and 3000 controls, significant risk factors for all stroke were: history of hypertension (OR 2.64, 99% CI 2.26-3.08; PAR 34.6%, 99% CI 30.4-39.1); current smoking (2.09, 1.75-2.51; 18.9%, 15.3-23.1); waist-to-hip ratio (1.65, 1.36-1.99 for highest vs lowest tertile; 26.5%, 18.8-36.0); diet risk score (1.35, 1.11-1.64 for highest vs lowest tertile; 18.8%, 11.2-29.7); regular physical activity (0.69, 0.53-0.90; 28.5%, 14.5-48.5); diabetes mellitus (1.36, 1.10-1.68; 5.0%, 2.6-9.5); alcohol intake (1.51, 1.18-1.92 for more than 30 drinks per month or binge drinking; 3.8%, 0.9-14.4); psychosocial stress (1.30, 1.06-1.60; 4.6%, 2.1-9.6) and depression (1.35, 1.10-1.66; 5.2%, 2.7-9.8); cardiac causes (2.38, 1.77-3.20; 6.7%, 4.8-9.1); and ratio of apolipoproteins B to A1 (1.89, 1.49-2.40 for highest vs lowest tertile; 24.9%, 15.7-37.1). Collectively, these risk factors accounted for 88.1% (99% CI 82.3-92.2) of the PAR for all stroke. When an alternate definition of hypertension was used (history of hypertension or blood pressure >160/90 mm Hg), the combined PAR was 90.3% (85.3-93.7) for all stroke. These risk factors were all significant for ischaemic stroke, whereas hypertension, smoking, waist-to-hip ratio, diet, and alcohol intake were significant risk factors for intracerebral haemorrhagic stroke. **INTERPRETATION:** Our findings suggest that ten risk factors are associated with 90% of the risk of stroke. Targeted interventions that reduce blood pressure and smoking, and promote physical

activity and a healthy diet, could substantially reduce the burden of stroke. FUNDING: Canadian Institutes of Health Research, Heart and Stroke Foundation of Canada, Canadian Stroke Network, Pfizer Cardiovascular Award, Merck, AstraZeneca, and Boehringer Ingelheim. Copyright 2010 Elsevier Ltd. All rights reserved.